Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. For this quiz, consider random variable X having Gaussian distribution with mean of 3 and variance of 4, random variable Y having Gaussian distribution with mean of -1 and variance of 9. That is, we can say that $X \sim \mathcal{N}(3, 4)$ and $Y \sim \mathcal{N}(-1, 9)$. Also consider random variable $Z = 2X + 3Y$.

   What is $\mathbb{E}[Z]$?

   Hint: The answers to all three questions on this quiz are all integers.

   1 / 1 point

   3

   Correct
   Yes. That is the correct answer.

2. Using X, Y, and Z from problem 1, if X and Y are uncorrelated, what is the variance of Z? That is, what is $\Sigma_{\tilde{Z}}$?

   1 / 1 point

   97

   Correct
   Yes. That is the correct answer.

3. Again using X, Y, and Z from problem 1, if X and Y are uncorrelated, what is $\mathbb{E}[Z \mid X = 1]$?

   1 / 1 point

   -1

   Correct
   Yes. That is the correct answer.